

ates; Primary contacts for system-related inquiries and administration.]
|IT Business Technology| Code Owners of Tech Stack; Advise Business and Technical Owners in supporting systems. |
|Security Risk|Help facilitate Tech Stack updates, including the Business Impact Analysis for new and existing systems.]

Tech Stack Definitions

- Business Owner: The Business Owner is the individual(s) responsible for all budget and decision making around the tool. They should define how the tool is used and by whom. This person(s) usually has login access to the tool as Owner but login access isn't necessary in all cases. Please make sure you list individual people in this field, rather than teams.
- Technical Owner: This may be the same as the Business Owner based on the nature of the system, for instance, if the system isn't centrally managed and or/requires little to no configuration. For systems that are centrally managed and/or heavily configured, the Technical Owner is someone with understanding of the application's architecture, functionality (including types of data involved), and overall criticality/impact to GitLab. Technical Owner(s) are typically administrators of a tool. This includes everyone with the administrative clearance to provision and deprovision access of a tool and/or as the technical expertise needed to manage it.
- Provisioner/Deprovisioner: People in charge of granting and removing team member access to a tool. At least two provisioners/deprovisioners should be named for every tool in the handbook. Provisioners can be contacted as a group in Slack using the @provisioners handle and in GitLab using the @tech-stack-provisioner handle.

Tech Stack Updates

We recommend using the Web IDE for Tech Stack Updates as we have a JSON Schema that will automatically validate your changes.

There are three main reasons you would want to change our Tech Stack:

- Adding a new system to the tech stack
- Updating the tech stack
 - This includes updating any of the columns in the tech stack, like Provisioner/Deprovisioner, Business Owner or Technical Owner.
- Removing a system from the tech stack

What data lives in the Tech Stack?

Please ensure that whenever you update the tech stack, you follow the instructions below and use the right type of data in each field.

Field	Type of data	Instructions	Responsible for filling out information
title	Text	The name of the system you are adding to the tech stack. Please enter the full and correct brand name associated with the tool. Example: "Zendesk, not ZenDesk."	MR Author and contributors
teammemberbaselineentitlement	Boolean	A baseline entitlement is a system that ALL GitLab Team Members get access to. Example: "Zoom". Most systems in our Tech Stack are not baseline	

entitlements. | MR Author and contributors |

| description | Text/Markdown | Business Purpose of the system. Please add links to handbook pages or websites that can provide people with more information on what the system is used for. Example: "ContractWorks is a contract managing software. This process is used to file contract or related vendor documents after they are fully executed." | MR Author and contributors |

| accessto | Text | Define which individuals or teams need access to this system. Example: Strategic Marketing and Product Managers | MR Author and contributors |

| provisioner | Text in single quotes, GitLab Username | Add the username of the people in charge of provisioning access to this system, separated by commas. We require at least 2 people to be listed as provisioners of a tool. Example: '@username, @username1' | MR Author and contributors |

| deprovisioner | Text in single quotes, GitLab Username | Add the username of the people in charge of removing access to this system, separated by commas. We require at least 2 people to be listed as deprovisioners of a tool. Team members can serve as both provisioners and deprovisioners of a tool. Example: '@username, @username1' | MR Author and contributors |

| groupowner | Text | Add the group owner of the tool (team/department/function). Example: Infrastructure | MR Author and contributors |

| groupownerslackchannel | Text | Add the Slack channel where the group owner can be reached out for help. Example: infrastructure-lounge | MR Author and contributors |

| businessowner | Text | The Business Owner is the individual(s) responsible for all budget and decision making around the tool. They should define how the tool is used and by whom. This person(s) usually has login access to the tool as Owner but login access isn't necessary in all cases. Please make sure you list individual people in this field, rather than teams. Example: Jane Doe, John Doe | MR Author and contributors |

| technicalowner | Text | The Technical Owner(s) all the administrators of a tool. This includes everyone with the administrative clearance to provision and deprovision access of a tool and/or as the technical expertise needed to manage it. Example: Jane Doe, John Doe. See guidance above for instances where a system does not require/have an administrator function | MR Author and contributors |

| dataclassification | Text (Red, Orange, Yellow, Green) or Unknown | Decided upon by the Security team, please leave as null while this process is completed. More information on Data Classification Standards. | Security Risk |

| authenticationmethod | Text (Okta SWA, Okta SAML, ID and password, other) or Unknown | Authentication method used to access the system. It can be SWA, SAML or other such as direct access (email and password login). | MR Author and contributors |

| criticalsystemstier | Text (Tier 1 Mission Critical, Tier 2 Business Critical, Tier 3 Business Operational, Tier 4 Administrative, TBD) or Unknown | This field classifies the system based on GitLab's Critical System Tier Definitions. The assignment of a critical system tier is dependent on the completion of a Business Impact Analysis (BIA) questionnaire. The Security Risk Team will coordinate the completion of a BIA if it has not yet been completed at the time a system is being added to the Tech Stack. | Security Risk |

| collecteddata | Text or Unknown | Data that is collected by the tool | MR Author and contributors |

| employeeorcustomerfacingapp | Text (employee, customer) | If access is limited to GitLab team members, then please add the employee word. If access can be granted to external parties, then add customer | MR Author and contributors |

| notes | Text or Unknown | Additional relevant information about the system that is not captured in any other field. Examples include the GitLab Epic for implementation and rollout. | Optional, MR Author and contributors |

| handbooklink | Text/Markdown or Unknown | Link to the Tech Stack Guide handbook page that

includes function and system information. Example: "The Marketing handbook contains the Marketo Tech Stack Guide." | Optional, MR Author and contributors |
externallink	Text/Markdown or Unknown	Link to the app's primary website. Example: "ContractWorks"	Optional, MR Author and contributors
googlegroup	Text or Unknown	Google group being used to manage access to the systems through Okta	Optional, MR Author and contributors
accessibility	Boolean	Does the application support accessibility features? Select Yes or No. (Existing applications have been temporarily updated with "TBD")	MR Author and contributor
accessibilitydescription	Text	Document instructions to enable or submit a request to activate the accessibility feature. (May include the link to a handbook page or request ingestion point)	MR Author and contributor
nowid	Text/ServiceNow ID	Integration ID for ServiceNow. Please leave blank for EntApps Engineer to Complete	Contributor, Service Now Engineer

- For Booleans, you need to type out either true or false.
- Unknown: If information is unknown, please don't leave the field empty, type null instead.

Add new system to the tech stack

We recommend using the Web IDE for Tech Stack Updates as we have a JSON Schema that will automatically validate your changes.

To add a new system to the tech stack, you must start a merge request in the tech stack yml file and follow the steps below.

Step 1

Copy the content below (including the - before title) and please add your system to the stack in its appropriate alphabetical placement. Make sure you fill out anything that the 'MR Author and contributors' are responsible for according to Tech Stack Data section. Make sure to use the correct data type for each required field.

yaml

```
- title:  
  teammemberbaselineentitlement:  
  description:  
  accessto:  
  provisioner:  
  deprovisioner:  
  groupowner:  
  groupownerslackchannel:  
  businessowner:  
  technicalowner:  
  dataclassification:  
  authenticationmethod:  
  criticalsystemstier: null  
  collecteddata: null  
  employeeorcustomerfacingapp:  
  notes: null
```

handbooklink: null
externallink:
googlegroup: null
accessibility:
accessibilitydescription:
nowid: null

Step 2

Submit MR and select the template tech-stack-add-new-system.md. Fill out all the information required in the description.

Step 3

The Legal, IT Compliance, Internal Audit, Security Risk, and Business Systems team will be automatically cc'd in the MR when you use the correct template. These teams will review and update the MR by filling out the missing fields relevant to their teams. Once it is all done, the Business Systems Analysts will merge.

Update Tech Stack Information

We recommend using the Web IDE for Tech Stack Updates as we have a JSON Schema that will automatically validate your changes.

To update any system information listed in the Tech Stack, you must start a merge request in the tech stack yaml file. Please use the tech-stack-update-existing-system.md template when submitting your MR and follow the instructions in the template. Ensure to update the information following the instructions in the Tech Stack Data section.

Removing a system from the Tech Stack

Occasionally systems listed in our tech stack will be deprecated. If a system is being offboarded (no longer being used and/or being replaced), please create an MR in the tech stack yaml file to remove the entry for the tool (DON'T MERGE IT), and then create an issue following this template. Once the issue has been submitted, link the MR in the comments. The Business Owner will need to work with Legal and IT Compliance to ensure that the data deletion process is being completed as outlined in the vendor contract regarding their process and responsibilities.

Other Related Processes

Access Requests

Please review our Access Requests handbook page to find more information on how to request access to systems. Choose the right template for your use case and follow the instructions.

Procurement

Do you want to procure a new application/system? Visit the Procurement handbook page to find

out more information on how to get started. Only applications which go through this process can be added to the tech stack.

Support

Are you experiencing issues with an application/system? Visit our Tech Stack YAML and under Group Owner/Slack Channel, you can find out what Slack channel you need to reach out to request support.

Compliance

Many applications listed in the Tech Stack are in-scope for regulatory or other compliance activities. For more information please reach out to the DRIs listed below:

- FedRAMP: Corey Oas (@corey-oas), Security Compliance
- GitLab SaaS SOC2: Liz Coleman (@lcoleman), Security Compliance
- GitLab Dedicated SOC2: Corey Oas (@corey-oas), Security Compliance
- ISO 27001: Liz Coleman (@lcoleman), Security Compliance
- SOX: Sean Brown (@sbrown10), Internal Audit

Updating the Offboarding Templates

The offboarding templates need to be updated when a new system is added to the Tech Stack in order for GitLab to be compliant and remove Team Members from systems once they leave GitLab. There are two different ways to update the offboarding templates. Pick the option which best suits the usage of the new system. Once the MR is ready to be merged, tag the People Connect team - @gl-people-connect-team - for approval and merge.

<details>

<summary markdown="span">Option 1: New system used by all/many Team Members, across multiple departments</summary>

Update the main Team Member Offboarding template.

The main template is divided into two sections. The first section is the Offboarding Tasks section. This section is applicable to all GitLab Team Members and is performed by the Manager, People, Business Technology, and Finance/Accounting teams.

The second section is the Tech Stack System Deprovisioning / Offboarding section. Update this section with the new system being added to the Tech Stack. Use the format below to make the update:

markdown

Department that owns the new system. If new system is owned by multiple departments, list the department the main deprovisioner belongs to. Please make sure the department does not already exist within the template. If it does, ignore this header and just add the following information underneath the existing department header

<summary>Name of deprovisioner @usernameofdeprovisioner, Name of deprovisioner 2 @usernameofdeprovisioner2, etc. </summary>

```

<table >
<tbody>
<tr>
<td> System name </td>
<td>

```

- [] Access Deprovisioned - System name
- [] Not Applicable to Team Member - System name

```

</td>
</tr>
</tbody>
</table>

```

When finished, link the MR in the "Access Tasks" section of the 'Tech Stack - Add New System' MR.

</details>

<details>

<summary markdown="span">Option 2: New system used by a limited number of Team Members, all in SAME department</summary>

Update the department-level Offboarding template within this directory.

Using the link above, locate the template of the department owning the new system and update it. Example: If only the Security team will have access to a new system, locate the Security template and add the new system to that template following this format:

- [] @usernameofdeprovisioner @usernameofdeprovisioner2: Remove the team member from System name

When finished, link the MR in the "Access Tasks" section of the 'Tech Stack - Add New System' MR.

</details>

Exceptions

There are exceptions to adding technologies to the Tech Stack which are handled on a case-by-case basis.

SECTION: business-technology/tech-stack.md

This is a listing of the applications we use at GitLab. If you're looking for the technology used to deliver GitLab.com, please visit the Production Architecture page.

Tech stack related processes

Update tech stack information

Please visit our Tech Stack main page for instructions on how to request updates to the Tech Stack information.

Request Access

If you need to request access for any of the following tools, please refer to and follow the Access Request process.

Tech Stack Applications

{{% all-tech-stack %}}

SECTION: business-technology/engineering/infrastructure/reference-links.md

<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />

Milestones

Each issue and epic gets mapped to the milestone for the time period that most of the work takes place. This is used as our lightweight version of time tracking and capacity planning, and is helpful with roadmap planning.

- Monthly Milestones
 - BT Infra 2021-07
 - BT Infra 2021-08
 - BT Infra 2021-09
 - BT Infra 2021-10
 - BT Infra 2021-11
 - BT Infra 2021-12
 - BT Infra 2022-01
- Quarterly Milestones
 - BT Infra FY22-Q3
 - BT Infra FY22-Q4
- Fiscal Year Milestones
 - BT Infra FY22-2H
 - BT Infra FY23-1H
 - BT Infra FY23-2H

Repositories

- gitlab.com/gitlab-com/business-technology/engineering/infrastructure - The group with all repositories for Business Technology Engineering Infrastructure that don't contain sensitive information that should be hosted on ops.gitlab.net.

Labels and Tags

Issue Types

You can view the issues with these labels on the BT Eng Infra - Issue Types issue board.

These labels can be applied to any issue, MR, or epic in the `gitlab.com/gitlab-com` namespace, which includes the majority of issue trackers used by GitLab team members in child groups and projects.

> <small>Note: These labels have not been created for the `gitlab-org` namespace since our team does not work on any projects in that namespace, and it would require the creation of additional issue boards and automation for the separate top-level group.</small>

<!-- This is an HTML table since multi-link links do not format well with Markdown tables. It cannot be indented due to Markdown indent formatting problems. -->

Label	When it's used	Links
<code>bt-infra::access-request</code>	Applied to all access requests that we responsible for provisioning.</small>	Issues Merge Requests Epics
<code>bt-infra::bug</code>	Applied to bug reports and quick fixes that isn't directly associated with a code repository that should be fixed ASAP. If this is a broader or longer lived issue, this should likely be refactored into another category and/or issues in the code repository issue tracker.</small>	Issues Merge Requests Epics


```

</td>
</tr>
<!-- bt-infra::epic-project -->
<tr>
<td><code>bt-infra::epic-project</code></td>
<td><small>Applied to larger scope or longer duration issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra%3A%3Aepic-project">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra%3A%3Aepic-project">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-infra%3A%3Aepic-project">Epics</a>
</td>
</tr>
<!-- bt-infra::iac -->
<tr>
<td><code>bt-infra::iac</code></td>
<td><small>Applied to Terraform, Ansible, and K8s changes that often require a certain headspace to complete. This allows us to resolve multiple issues simultaneously.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra%3A%3Aiac">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra%3A%3Aiac">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-infra%3A%3Aiac">Epics</a>
</td>
</tr>
<!-- bt-infra::interest -->
<tr>
<td><code>bt-infra::interest</code></td>
<td><small>Applied to issues that we're watching or are interested in as an FYI.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra%3A%3Ainterest">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra%3A%3Ainterest">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-infra%3A%3Ainterest">Epics</a>
</td>
</tr>
<!-- bt-infra::service -->
<tr>
<td><code>bt-infra::service</code></td>
<td><small>Applied to issues related to providing (internal) customer service and/or for issues relating to one of the managed services that we offer. If we're helping people and it's not a bug, this label is usually applied.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-

```

```

infra%3A%3Aservice">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-
infra%3A%3Aservice">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-
infra%3A%3Aservice">Epics</a>
</td>
</tr>
<!-- bt-infra::unplanned -->
<tr>
<td><code>bt-infra::unplanned</code></td>
<td><small>Applied to any asks from team members or other departments that are not part of our
day-to-day responsibilities or managed services that we offer.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-
infra%3A%3Aunplanned">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-
infra%3A%3Aunplanned">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-
infra%3A%3Aunplanned">Epics</a>
</td>
</tr>
<!-- bt-infra::wishlist -->
<tr>
<td><code>bt-infra::wishlist</code></td>
<td><small>Applied to our long-lived backlog of things that are relevant and should not be
rejected, however are not a priority right now.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-
infra%3A%3Awishlist">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-
infra%3A%3Awishlist">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/epics?labelname%5B%5D=bt-
infra%3A%3Awishlist">Epics</a>
</td>
</tr>
</table>

```

Managed Services

You can view the issues with these labels on the BT Eng Infra - Managed Services issue board.

<!-- This is an HTML table since multi-link links do not format well with Markdown tables. It cannot be indented due to Markdown indent formatting problems. -->

```

<table>
<thead>
<tr>
<th style="width: 35%;">Label</th>
<th style="width: 40%;">When it's used</th>
<th style="width: 25%;">Links</th>
</tr>

```

```

</thead>
<tbody>
<!-- bt-infra-service::aws -->
<tr>
<td><code>bt-infra-service::aws</code></td>
<td><small>Applied to all AWS infrastructure related issues that are not specific to the GitLab
Sandbox Cloud.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Aaws">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Aaws">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Aaws">Epics</a>
</td>
</tr>
<!-- bt-infra-service::demo-systems -->
<tr>
<td><code>bt-infra-service::demo-systems</code></td>
<td><small>Applied to all Customer Success Demo Systems related issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Ademo-systems">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Ademo-systems">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Ademo-systems">Epics</a>
</td>
</tr>
<!-- bt-infra-service::dns -->
<tr>
<td><code>bt-infra-service::dns</code></td>
<td><small>Applied to all DNS domain, zone, and record related issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Adns-zones">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Adns-zones">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Adns-zones">Epics</a>
</td>
</tr>
<!-- bt-infra-service::gcp -->
<tr>
<td><code>bt-infra-service::gcp</code></td>
<td><small>Applied to all Google Cloud Platform (GCP) issues that are not specific to the
GitLab Sandbox Cloud.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Agcp">Issues</a><br />

```

```

<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Agcp">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Agcp">Epics</a>
</td>
</tr>
<!-- bt-infra-service::infra-standards -->
<tr>
<td><code>bt-infra-service::infra-standards</code></td>
<td><small>Applied to all Infrastructure Standards and Infrastructure shared services
issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Ainfra-standards">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Ainfra-standards">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Ainfra-standards">Epics</a>
</td>
</tr>
<!-- bt-infra-service::platypus -->
<tr>
<td><code>bt-infra-service::platypus</code></td>
<td><small>Applied to all BT Enterprise Apps Integrations Project Platypus infrastructure
related issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Aplatypus">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Aplatypus">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Aplatypus">Epics</a>
</td>
</tr>
<!-- bt-infra-service::sandbox-cloud -->
<tr>
<td><code>bt-infra-service::sandbox-cloud</code></td>
<td><small>Applied to all GitLab Sandbox Cloud and HackyStack issues.</small></td>
<td>
<a href="https://gitlab.com/groups/gitlab-com/-/issues?labelname%5B%5D=bt-infra-
service%3A%3Asandbox-cloud">Issues</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Asandbox-cloud">Merge Requests</a><br />
<a href="https://gitlab.com/groups/gitlab-com/-/mergerequests?labelname%5B%5D=bt-infra-
service%3A%3Asandbox-cloud">Epics</a>
</td>
</tr>
</table>

```

SECTION: business-technology/entapps-documentation/_index.md

SECTION: business-technology/entapps-documentation/policies/_index.md

<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />

<i class="fas fa-users" id="biz-tech-icons"></i> Links to GitLab Policies

1. <<https://handbook.gitlab.com/handbook/security/security-and-technology-policies/access-management-policy/>>
2. <<https://handbook.gitlab.com/handbook/security/standards/password-standard/>>
3. <<https://handbook.gitlab.com/handbook/security/security-and-technology-policies/change-management-policy/>>
4. <<https://handbook.gitlab.com/handbook/security/security-and-technology-policies/audit-logging-policy/>>
5. <<https://handbook.gitlab.com/handbook/business-technology/entapps-documentation/policies/gitlab-business-continuity-plan/>>
6. <<https://handbook.gitlab.com/handbook/engineering/incident-management/>>
7. <<https://handbook.gitlab.com/handbook/security/security-operations/sirt/sec-incident-response/>>

SECTION: business-technology/entapps-documentation/policies/gitlab-business-continuity-plan.md

Purpose

Business Continuity Plan is the process involved in creating a system of prevention and recovery from potential threats to GitLab. The plan ensures that personnel and assets are protected and are able to function quickly in the event of a disaster.

Scope

GitLab, by its remote-only nature, is not easily affected by typical causes of business disruption, such as local failures of equipment, power supplies, telecommunications, social unrest, terrorist attacks, fire, or natural disasters. System data from the Business Impact Analysis may be leveraged as part of business continuity planning and testing. Additionally, the BCP works in conjunction with the Disaster Recovery Plan (DRP).

Roles & Responsibilities

| Role | Responsibility |

|-----|-----|

| GitLab Team Members | Responsible for following the requirements in this procedure |

Business Technology	Responsible for implementing and executing this procedure
Business Technology Management (Code Owners)	Responsible for approving significant changes and exceptions to this procedure
Security Risk	Responsible for conducting periodic Business Impact Analysis and applying Critical System Tiers to Tech Stack Systems

Procedure

BCP for Remote Workers

In case of an all-remote company like GitLab, it is sufficient to have simple contingency plans in the form of service-level agreements with companies that host our data and services. The advantage of an all-remote workforce like GitLab is that if there are clusters of people or systems that are unavailable, the rest of the company will continue to operate normally.

The exception to this would be a scenario of a single point of failure, (for example, if one of the Engineering heads who should sign off on triggering the plan is unavailable due to a disaster). In this case we would need an alternate plan in place that covers how to get in contact with the person or people affected by the disaster and trigger this business continuity plan.

Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

RTO and RPO are two of the most important parameters of a Business Continuity Plan. These are objectives to guide GitLab Infrastructure team in choosing the optimal data backup plan. The RTO/RPO provides the basis for identifying and analyzing viable strategies for inclusion in the business continuity plan. Viable strategy options include any which would enable resumption of a business process in a time frame within the RPO/RTO.

What is a Recovery Point Objective?

Recovery Point Objective (RPO) is the interval of time that might pass during a disruption before the quantity of data lost during that period exceeds the Business Continuity Plan's maximum allowable threshold.

What is a Recovery Time Objective?

The Recovery Time Objective (RTO) is the duration of time a service level or business process must be restored after a disaster, in order to avoid risks associated with a break in continuity.

What triggers the Business Continuity Plan

For a business continuity plan to be effective, it needs to be triggered as soon as possible; too early or late can reduce its efficacy. Key decision points to consider when a BCP has to be triggered or invoked are given below:

- When an incident turns into an event like a disaster, breach, or something which classifies as a Severity 1
- When the estimated time of resolution for a potential breach is greater than the normal

estimated time for regular security incidents

- When the recovery of an incident is uncertain, a decision must be made to invoke the business continuity plan if the disruption cannot be resolved within the specified incident recovery timelines
- When resolution of an incident with critical customers, depending on their service-level agreements is delayed, then the BC plan must be triggered

Data Continuity System

This section provides details about the production environment that must be available for GitLab.com (for both SaaS services Dedicated and Commercial) to run effectively:

GitLab.com and customers.gitlab.com are hosted on Google Cloud Platform. Dedicated is hosted on AWS.

Priority::1: Outage would have immediate impact on GitLab customer/user operations

1. Disruption of service of Google Cloud Platform and AWS, specifically the region in which GitLab.com, dev.gitlab.org and ops.gitlab.net are hosted (for both Dedicated and GitLab.com).

- Effect: a loss or degradation of service, of either Google Cloud Platform or AWS means that GitLab.com or GitLab Dedicated are not available. This affects anyone who uses GitLab.com or GitLab Dedicated to host their repositories and may prevent GitLab team members from completing their work. GitLab.com is also the primary server where GitLab CE and EE source code and packages are hosted.

- Solution(s): Ensure that ops.gitlab.net and dev.gitlab.org mirror critical repositories necessary for operating GitLab.com and GitLab Dedicated, and that they are running from different regions. Ensure that there are many other servers across the globe where GitLab CE/EE is readily available.

- Effect: Security releases are developed on GitLab.com and staged on dev.gitlab.org before being brought to production on GitLab.com; Parts of these may be lost or unavailable for the duration of the disruption.

- Solution(s): Depending on the duration and nature of the disruption, the solution is to wait for service to be restored (minimal duration), or build a new set of servers. Leverage database snapshots, and initiate recovery from backup as it is relatively quick.

1. Unavailability of support staff in case of a customer emergency.

- Effect: emergency response times are greater than intended.

- Solution(s): The team is distributed geographically (except during team get-togethers). Customer emergencies are handled by any person who is in the on-call rotation. The on-call load is distributed at many levels, service engineers, production engineers, and even developers can be summoned when we have an outage or a customer incident. Emergencies also trigger automatic notifications on our internal chat system, alerting the entire company. There is also an ongoing effort to publish our runbooks, explaining how we manage our infrastructure and how we deal with outage cases.

1. Disruption of service of ZenDesk.

- Effect: support workflows are disrupted. New tickets cannot be created, existing tickets cannot be responded to.

- Solution(s): If the outage has lasted for more than 4 hours, Support will update the GitLab status page to indicate the support system is down and

to direct all requests to a temporary inbox. Support agents will work via this inbox for the duration of the outage. Agents will log the emails worked. After service is restored, Support Readiness will work to get the data from the emails back into the support system.

P2: Outage would have immediate impact on GitLab ability to continue business
Malicious Software attack and hacking or other Internet attacks.

- Effect: depends on attack.
- Solution(s): We log and track any access that happens on any server in the fleet using logstash/kibana at log.gprd.gitlab.net.

P3: Outage greater than 72 hours would have impact on GitLab ability to continue to do business
Disruption of service from Salesforce.com, Zuora, NetSuite, Google Workspace

- No failover plan currently.

P4: Non critical system
Disruption of service from TripActions or internal chat tool (Slack).

- When TripActions is down, team members can use their own travel booking tool and expense it to GitLab with reason for exception
- When Slack is down, team members can use Zoom Chat as outlined here.

Communication Plan and Role Assignments

When it comes to a disaster, communication is of the essence. A plan is essential because it puts all team-members on the same page and clearly outlines all communication. Documents should all have updated team-member contact information and team-members should understand exactly what their role is, in the days following the triggering of the BC plan. Assignments like setting up workstations, assessing damage, redirecting phones and other tasks will need assignments if you don't have some sort of technical resource to help you sort through everything.

Each GitLab team should be trained and ready to deploy in the event of a disruptive situation requiring plan activation. The plan of action steps, procedures, and guidelines will be documented in their team runbooks page (currently under development) and should be available offline. This should have detailed steps on recovery capabilities, and instructions on how to return the system to normal operations.

More details on this will be covered in the BC plan - roles & responsibilities section which is in development.

Backup check

Make sure that backups are performed daily, and include running an additional full local backup on all servers and data in the Business Continuity preparation plan. Run them as far in advance as possible to ensure that they're backed up to a location that will not be impacted by the disaster. Alternate storage provisioning.

Distribute and Verify the Plan / Approval from Senior management

- GitLab documentation related to all procedure and guidelines detailing the Business Continuity and Disaster Recovery must be reviewed, updated, and formally approved by GitLab leadership
- After developing basic guidelines, this plan will be distributed as a work in progress to the core team
- The core team will review to verify that all technical details are covered and deficiencies exist
- The core team will review and add guidelines so that all related DRIs are clear on what is required in all situations
- GitLab team members must be asked to confirm their current details, such as phone numbers and emergency contacts on an annual basis
- Managers or team leads will share the relevant parts of the plan to all GitLab team members based on their role and department, and verify that they know what to do in the event of an emergency

Vendor communication and service restoration plan

A plan cannot be successful without restoring customer confidence. As a final step, ensure that there is a detailed vendor communication plan as part of the Business continuity preparation plan. This plan will check for all the systems and services to ensure normal operations have resumed as intended once the damage is repaired in the area. Also, include the section to check with the main service providers on restoration and access.

Root Cause Analysis

Any time the business continuity plan is activated, a root cause analysis should be performed to identify lessons-learned. The root cause analysis should review the trigger of the event and recommend remediations that prevent future occurrences of the issue. Additionally, if opportunities for improvements in the response to the specific business continuity scenario are identified, the business continuity plan and applicable procedures should be updated to reflect those lessons learned.

Business Continuity Test

After formalizing the business continuity plan, or BCP, the next important step is to test the plan. Testing verifies the effectiveness of the plan, trains plan participants on what to do in a real scenario, and identifies areas where the plan needs to be strengthened. A test of the plan review, has to be conducted at least annually.

GitLab's first test of the business continuity plan was performed in April 2020 and tests will be conducted at least annually or when significant business changes occur.

Why Business continuity testing is important

- To ensure that the current backup facilities and procedures are feasible and compatible to achieve the determined RTO. Can backup systems withstand a cyberattack.
- To confirm that your continuity objectives are met (RTO and RPO). Accordingly provide training to the team managers and team members.

- To evaluate the company's response to various kinds of disruptive events .. Emergency communication strategy, is it functioning as expected. - How quickly can everyone be informed about an incident.
- To identify areas in the plan that need modification. Improve systems and processes based on test findings. And accordingly maintain and update the BC plan.

Testing the plan

Testing can present a lot of challenges. It requires investing time and resources. With that in mind, to start with, it may make more sense to conduct a tabletop test at a conference room, rather than involving the entire organization in a full-blown drill.

Also an initial "dry run" of the plan can be performed, by conducting a structured walk-through test of the approved BC plan. The initial testing is done in sections and after normal business hours to minimize disruptions.

Subsequent tests can occur during normal business hours. An actual test-run can be performed eventually.

Based on the gaps and weaknesses learnt from the testing, underlying problems should be corrected and the plan updated accordingly.

The various types of tests that can be conducted include: checklist tests, simulation tests, parallel tests, and full interruption tests

Not testing the plan will put both the business and customer confidence at risk.

Business Continuity Plan Testing Scenarios

There are several types of tests, such as a plan review, a tabletop test, or a simulation test, which was detailed in the previous section.

Some testing scenarios that can be performed, are given below:

1. Data Loss/Breach

- One of the most prevalent workplace disasters today. Cause of data loss or breach could be due to any of the following:

- Ransomware and cyberattacks
- Unintentionally erased files or folders
- Server/drive crash
- Datacenter outage

- Data is mission-critical and losing it can have many serious consequences, such as significantly impacting sales and logistics applications.

- The goal is to regain access to that data as soon as possible. Restoring backup is the solution. However, who's responsible for that? What's the communication plan in this case? What are the priorities? Who needs to be contacted right away? Are there any vendors involved? These and many other questions will be answered during this test.

1. Data Recovery Testing

- This testing scenario, is used to make sure that the backup and recovery systems work as intended. To prove that, run a test that involves losing a bulk of data, and then try to recover it.

- Some of the elements to be evaluated include the RTO, and whether the team met its objectives.

- Also make note of, if there were any damage to the files during recovery? Was the backup stored in the cloud, recovered successfully and on time.

1. Emergency Communication

- Being able to communicate during a disaster or an emergency is crucial. Yet, the most disruptive events can leave with no traditional means of staying in contact.

- For these scenarios, the BC plan needs to outline the actions to be taken. An alternate mode of communication should be tested for its reliability and efficiency, for a company like GitLab which has team members all around the globe.

- Regular updates to all GitLab team members contact information, so that all of them can receive timely notifications thus streamlining the disaster scenario process.

Exceptions

Exceptions to this procedure will be tracked as per the Information Security Policy Exception Management Process.

References

- Business Impact Analysis
- Disaster Recovery Plan (DRP)

SECTION: business-technology/entapps-documentation/procedures/_index.md

<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />

<i class="fas fa-users" id="biz-tech-icons"></i> Links to IT Enterprise Applications Procedures

Please note this page is a work in progress

1. GitLab Team Member How to Guides

1. Standard Operating Procedures

SECTION: business-technology/enterprise-applications/_index.md

<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />

<i class="fas fa-users" id="biz-tech-icons"></i> Enterprise Applications Divisions

The IT Enterprise Applications team oversees the business technology architecture and systems that power GitLab as a business entity. We ensure that the systems are fit-for-purpose, available, reliable, integrated and secure. We continue to build business aligned technology to run operations, layering integration and automations to support remote work, enhance team member and customer experience, and generate value.

Our team members bring diverse expertise to ensure we deliver high-quality solutions and support for our internal customers. Please refer to the directory listing below to review the individual divisions, policies, and procedures that encompasses IT Enterprise Applications

SECTION: business-technology/enterprise-applications/entapps-crm/_index.md

<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />

<i class="fas fa-bullseye" id="biz-tech-icons"></i> Mission

Our mission is to deliver a robust, end-to-end solution that streamlines and enhances the customer lifecycle at every stage, including:

- Lead Generation and Management: Providing tools and automations that drive efficient lead capture, nurturing, and conversion.
- Sales Enablement: Optimizing sales cycles through simplified workflows and actionable insights.
- Subscription Management: Ensuring a seamless quote-to-cash process with Zuora's subscription and billing capabilities.
- Real-Time Insights: Delivering data-driven insights that empower teams to make informed decisions and strengthen customer relationships.

We Aim to

- Automate and Optimize: Eliminate manual bottlenecks, ensuring processes are efficient and error-free.
- Scale with the Business: Build systems capable of supporting GitLab's evolving needs as we grow.
- Empower Teams: Equip internal teams with tools and knowledge to build stronger customer connections, increase operational efficiency, and accelerate growth.

<i class="fas fa-users" id="biz-tech-icons"></i> Our Team

Learn more about our team and roles in supporting GitLab's enterprise applications.

<i class="fas fa-bullhorn" id="biz-tech-icons"></i> What We Do

System Integration and Optimization

- Seamlessly integrate Salesforce and Zuora, ensuring cohesive operations between CRM and subscription management platforms.
- Develop scalable solutions that support global sales, marketing, and customer success teams.

Automating Workflows

- Create automation that eliminates redundant tasks, accelerates sales cycles, and enhances data accuracy across systems.

Enhancing the Customer Lifecycle

- Enable GitLab teams to deliver a superior customer experience through optimized lead management, seamless subscription workflows, and precise billing operations.

Delivering Insights

- Build reporting and analytics capabilities that provide real-time visibility into customer behaviors, revenue trends, and operational metrics.

How we Operate

Our team uses a flexible intake model to manage requests of varying complexity. This ensures that we can address both quick fixes and large-scale, multi-team projects efficiently.

Operational Bugs or General Questions

For any operational bugs or general inquiries related to CRM systems, please submit a request via:

- HelpLab - IT Break / Fix Request

Small to Medium Feature Requests & Enhancements

For smaller feature requests or system enhancements:

Weekly Intake Process:

- Participate in our weekly intake meetings to present your request.
- Each department has designated representatives who will help channel requests effectively. You can reach out to them directly:

1. Sales: RSOE
2. Marketing: Marketing Operations
3. Other Departments: Please reach out to the appropriate point of contact.

Salesforce or Zuora Issues:

- If your request is related to Salesforce or Zuora, please choose the "CM: Configuration Change (Generic)" template when submitting.

Large and Complex Projects

For larger, cross-functional projects requiring significant effort and resources:

Quarterly PMO Intake Process:

- For projects that span multiple teams or involve extensive development work, use the quarterly intake process to submit requests. This process ensures that projects are scoped, prioritized, and allocated resources appropriately.

Our Commitment

- Strengthen Customer Relationships: Equip teams with tools that enable them to anticipate and exceed customer expectations.
- Enhance Operational Efficiency: Reduce time and resources spent on manual processes by implementing smart automation and integrations.
- Accelerate Revenue Growth: Support GitLab's sales and finance teams with systems that ensure accurate and timely billing, seamless subscription management, and efficient deal closures.
- Drive Innovation: Continuously iterate and adopt industry best practices to maintain a cutting-edge CRM and billing ecosystem.

SECTION: business-technology/enterprise-applications/entapps-crm/finsys-access-requests.md

Access Requests

Many finance systems that are provisioned are also considered a SOX system and are subject to the SOX Program.

The finance systems admin team goes through additional procedures when handling Access Requests.

This applies to the following systems:

- Adaptive Insights
- Avalara
- Coupa
- Navan
- NetSuite
- Stripe
- Zuora Billing
- Zuora Revenue

Additional Procedures

On top of the GitLab Access Request process, the team will perform the following:

- Ensure/Clarify access being requested is specific to the system being requested.
 - Example: Instead of Needs read only access to Netsuite, the team will clarify with the requester that they are asking for the Custom Auditor (read only) role in Netsuite.
 - See matrix here for System specific requirements.
- Seek Business Owner's approval of the exact role and/or permission being requested, according to who is listed in the Tech Stack. This includes baseline entitlement(s).
- Once approved, the team will provision the user in the system and add to the corresponding Okta Google group (if necessary).
- Once provisioned, the team will take a screenshot of the user record in the system with a timestamp and attach it to the issue and then close.

Note: Unless the AR specifically mentions an environment besides Production, the team will

always assume the requester is only requesting access for the Production environment.

Definition of Done

- An AR is only considered Done from the FinSys perspective when it has the following:
 - Clear system specific roles and permissions being requested
 - Manager AND Business Owner Approver
 - Screenshot with timestamp of provisioned user record

System Specific Access Request Requirements

System	AR Requirements	Notes
Adaptive Insights	AR should explicitly list: Permissions Set (one):	James Shen (Sr. Director of FP&A) should approve every single AI AR.
	AI is set up with various 'Levels' which essentially corresponds to our Departments, so most users should not/will not have access to all the levels, since AI houses a lot of confidential data.	
	- Administrative	
	- Analysis	
	- Editable Sheet Access Only	
	- IT Administrator	
	- Report Only	
	- Standard	
	Levels:	
	- Listed out.	
	Be extremely cautious with levels. If Master Corporate or Corporate checkbox is selected, the requester will inherently have access to everything, which is unlikely what was intended.	
Avalara	AR should explicitly state the Exact role they are requesting access to.	Roles are not customizable.
	Roles:	
	- Account administrator	
	- Company administrator	
	- Read-only account access	
	- Read-only company access	
	- No access	
Coupa	Note: Requestors/buyers will always only get Custom User role. Elevated access requires Procurement approval.	
Navan	AR should explicitly state the Exact role they are requesting access to AND which policies. There are a lot of policies within Navan. Typically the only ARs the team will get are for Navan Admins. Team members are automatically provisioned.	
	Since most of the ARs are for finance system admins, they will usually need access to all the policies in which case the team does not have to list out all the policies in the AR.	
NetSuite	AR should explicitly state the Exact NetSuite role that they are requesting before the Business Owner approves the AR.	
Stripe	AR should explicitly state the Exact role they are requesting access to AND which Stripe Accounts.	
	- about.gitlab	

	- Bizzabo		
	- GitHub		
	- Zoom		
	Roles:		
	- Administrator		
	- Analyst		
	- Developer		
	- Support Specialist		
	- View Only		

| Zuora Billing | AR should explicitly list the role they are requesting access to for EACH module: |

| | - Platform |

| | - Billing |

| | - Payments |

| | - Finance |

| | - Commerce |

| | - Reporting |

| Zuora Revenue | AR should explicitly state the Exact ZREV role(s) they are requesting before the Business Owner approves the AR. |

SECTION: business-technology/enterprise-applications/entapps-crm/quote-to-cash/_index.md

Lead to Cash Flow

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```

```

end: 5;">Fulfillment Engineers</div>
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<div class="grid-item-small" style="background-color: 37b96d; grid-column-start: 3;grid-column-
end: 4;">Order Form Renewal (SFDC, Z-CPQ, DocuSign)</div>
<div class="grid-item-small" style="background-color: 168f48; grid-column-start: 4;grid-column-
end: 5;">See Sell</div>
</div>
</div>

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Process Flow Diagrams

Sales-Assisted: New Subscription

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Sales-Assisted: Renewal

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Zero Touch: New Subscription

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Entity Relationship Diagrams

- Zuora Billing
- CustomersDot (Customers Portal)
- GitLab

Quote to Cash Systems Field Mapping

This spreadsheet provides a mapping of fields from the data models of Zuora Billing, CustomersDot and GitLab. This document highlights some of the overlap and dependencies between the systems currently.

Contact Us

Slack Channels

- enterprise-apps
- business-technology
- bt-finance-operations
- financesystemshelp
- bt-integrations

GitLab Issues

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content:center;align-items:center;">Business Systems Analysts</a>
  <a href="https://gitlab.com/gitlab-com/business-technology/enterprise-
apps/integrations/issue-tracker/-/issues/new" class="btn btn-purple-inv"
style="width:33%;height:100%;margin:1px;display:flex;justify-content:center;align-
items:center;">Integrations Engineering</a>
  <a href="https://gitlab.com/gitlab-com/business-technology/enterprise-
apps/financeops/finance-systems" class="btn btn-purple-inv"
style="width:33%;height:100%;margin:1px;display:flex;justify-content:center;align-
items:center;">Finance Systems</a>
```

SECTION: business-technology/enterprise-applications/entapps-crm/quote-to-cash/entity-relationship-diagrams.md

```
<link rel="stylesheet" type="text/css" href="/stylesheets/biztech.css" />
```

Quote to Cash Entity Relationship Diagrams

Zuora Billing

mermaid
erDiagram

```
Product ||..|{ ProductRatePlan : contains
ProductRatePlan ||..|{ ProductRatePlanCharge : contains
ProductRatePlanCharge ||..|{ ProductRatePlanChargeTier : contains
Account ||..o{ Subscription : has
Subscription ||..|{ RatePlan : has
RatePlan ||..|{ RatePlanCharge : has
RatePlanCharge ||..|{ RatePlanChargeTier : has
ProductRatePlan ||..|{ RatePlan : "catalog trace"
ProductRatePlanCharge ||..|{ RatePlanCharge : "catalog trace"
Subscription ||..o{ Amendment : has
Account ||..|{ Contact : has
Account ||..o{ PaymentMethod : has
```

Account ||..o{ PaymentMethodSnapshot : has
 PaymentMethod ||..|| PaymentMethodSnapshot : has
 Account ||..o{ Invoice : has
 Account ||..o{ Payment : has
 Invoice ||..||{ InvoiceItem : contains
 InvoiceItem ||..|| RatePlanCharge : "subscription trace"
 Invoice ||..o{ InvoicePayment : "payment trace"
 InvoicePayment ||..|| Payment : "payment trace"
 InvoiceItem ||..o{ TaxationItem : tax
 Account ||..o{ Refund : has
 Refund ||..|| PaymentMethodSnapshot : source
 Refund ||..o{ RefundInvoicePayment : "refund trace"
 RefundInvoicePayment ||..|| InvoicePayment : "refund trace"
 InvoiceItem ||..o{ Amendment : source
 InvoiceItem ||..|| Subscription : "subscription trace"
 Account {
 varchar accountnumber
 varchar additionalemailaddresses
 boolean allowinvoiceedit
 boolean autopay
 decimal229 balance
 varchar batch
 varchar bcdsettingoption
 bigint billcycleday
 varchar-fk billtoid
 varchar communicationprofileid
 varchar-fk createdbyid
 timestampwithtimezone createddate
 decimal229 creditbalance
 varchar creditmemotemplateid
 varchar-fk crmid
 varchar currency
 varchar customerservicerepname
 varchar-fk debitmemotemplateid
 varchar-fk defaultpaymentmethodid
 varchar-pk id
 boolean invoicedeliveryprefsemail
 boolean invoicedeliveryprefssprint
 varchar invoicetemplateid
 date lastinvoicedate
 decimal229 mrr
 varchar name
 varchar notes
 varchar parentid
 varchar paymentgateway
 varchar paymentterm
 varchar purchaseordernumber
 varchar salesrepname
 varchar sequencesetid
 varchar-fk soldtoid

varchar status
varchar taxcompanycode
varchar taxexemptcertificateid
varchar taxexemptcertificatetype
varchar taxexemptdescription
date taxexempteffectivedate
varchar taxexemptentityusecode
date taxexemptexpirationdate
varchar taxexemptissuingjurisdiction
varchar taxexemptstatus
decimal229 totaldebitmemobalance
decimal229 totalinvoicebalance
decimal229 unappliedbalance
decimal229 unappliedcreditmemoamount
varchar-fk updatedbyid
timestampwithtimezone updateddate
varchar vatid
varchar acc